Scot Travels Rate Matrix Calculation Definition

This document defines only the rate calculation logic for the state-based rate matrix. It is aligned to the current implementation and is intended for admin, QA, and future development reference.

Version: 1.0
Scope: State-based vehicle rates, progressive slab pricing, surcharges, tax, and quote total calculation
Currency: GBP

1. Purpose

The rate matrix controls how a quote total is calculated for a specific pickup route, state, and vehicle. Each active state can have separate rates for each active vehicle. The quote engine selects the correct state, then calculates one fare option per configured vehicle inside that state.

Screenshot Placeholder 1
Add screenshot of the admin rate matrix screen here.

2. Calculation Scope

- Rate selection is state-specific.
- Vehicle pricing is state-specific.
- Distance pricing uses progressive slabs for all vehicles.
- Meet & greet, airport pickup, waypoint fee, night surcharge, and tax are added on top.
- For return journeys, each leg is calculated separately and then summed.

3. Rate Record Structure

Each rate record is unique for one combination of **state + vehicle**.

FIELD	TYPE	PURPOSE
stateId	Reference	The state this rate belongs to.
vehicleId	Reference	The vehicle this rate belongs to.
minimumBillableMiles	Integer	Minimum miles to charge even if actual route miles are lower.
slabs[]	Array	Progressive mileage slabs used to calculate distance charge.
airportPickupFee	Number	Flat fee added to every quote using this rate.
meetAndGreetFee	Number	Flat fee added only when meet & greet is selected.
waypointFee	Number	Flat fee applied per stop/waypoint.
nightWindowStart	HH:MM	Night charging window start time.
nightWindowEnd	HH:MM	Night charging window end time.

FIELD	TYPE	PURPOSE
nightSurchargeFee	Number	Flat night surcharge added when pickup falls inside night window.
taxRate	Number	Tax percentage stored as decimal. Example: 0.20 = 20%.
active	Boolean	Only active rates are used in quote calculation.

4. State Resolution Rule

1. Only active states are considered.
2. The pickup location decides the state.
3. A state matches if one of these is true:

- pickup.state exactly matches the state code
- pickup.state exactly matches the state name
- pickup.address contains the state name

If only one active state exists and no exact match is found, that single active state is used. If multiple active states exist and none matches, quote generation fails.

5. Vehicle Rate Selection Rule

1. Once the state is resolved, only rates for that state are loaded.
2. Only active rates are loaded.
3. Only rates whose linked vehicle is also active are shown.
4. Fare options are sorted by vehicle display order, then by vehicle name.
5. There can be only one rate record for one state + one vehicle.

Screenshot Placeholder 2
Add screenshot of the state-specific vehicle rate table here.

6. Progressive Slab Rules

Slabs are validated before saving. The following rules are enforced:

- At least one slab is required.
- The first slab must start from **1 mile**.
- `fromMiles` must be an integer greater than or equal to 1.
- `toMiles` must be null or an integer greater than or equal to `fromMiles`.
- `ratePerMile` must be greater than 0.
- Slabs must be contiguous with no gaps and no overlaps.
- Only the last slab may be open-ended (`toMiles = null`).
- Slabs are normalized and stored in order.

Sample Slab Configuration

```
[
  { "fromMiles": 1,  "toMiles": 10,  "ratePerMile": 4.0 },
```

```
  { "fromMiles": 11, "toMiles": null, "ratePerMile": 3.5 }
]
```

Meaning: the first 10 billable miles are charged at 4.00 per mile. Any billable miles above 10 are charged at 3.50 per mile.

7. Input Values Used in Fare Calculation

INPUT	SOURCE	NOTES
miles	Google route metrics	Actual route miles after pickup, dropoff, and stops are calculated.
stops	Journey builder	Waypoint fee depends on the count of stops.
includeMeetAndGreet	Booking details form	Controls whether meet & greet fee is added.
pickupAtLocal	Journey date/time	Used only to test night surcharge eligibility.
rate	StateVehicleRate	Includes all fees, slabs, tax, and minimum billable miles.

8. Fare Calculation Formula

Step 1: Billable Miles

```
billableMiles = max(actualMiles, minimumBillableMiles)
```

Step 2: Distance Charge Using Progressive Slabs

Distance charge is not flat per mile across the whole trip. It is calculated progressively slab by slab.

```
distanceCharge = sum(each covered slab miles × slab ratePerMile)
```

Step 3: Optional and Flat Add-ons

```
airportPickupFee = configured flat fee
meetAndGreetFee = configured fee if meet & greet is selected, otherwise 0
waypointFee = numberOfStops × configured waypoint fee
```

Step 4: Base Subtotal

```
baseSubtotal = distanceCharge + airportPickupFee + meetAndGreetFee + waypointFee
```

Step 5: Night Surcharge Check

The pickup time is checked against the configured night window. If the night window crosses midnight, the logic still works correctly.

- If start time equals end time, night surcharge applies all day.
- If start time is earlier than end time, pickup must be within the same-day window.
- If start time is later than end time, the window crosses midnight.

```
nightSurchargeApplied = pickupAtLocal falls inside configured night window
surchargeTotal = nightSurchargeApplied ? nightSurchargeFee : 0
```

Step 6: Tax and Final Total

```
subtotal = baseSubtotal + surchargeTotal
taxAmount = subtotal × taxRate
total = subtotal + taxAmount
```

All currency values are rounded to 2 decimal places before returning the final breakdown.

9. Worked Example

Example assumptions:

- Actual miles: 12.00
- Minimum billable miles: 3
- Slabs: 1-10 = 4.00 per mile, 11+ = 3.50 per mile
- Airport pickup fee: 8.00
- Meet & greet selected: Yes, fee 12.00
- Stops: 2, waypoint fee 6.00 each
- Night surcharge applies: Yes, fee 20.00
- Tax rate: 0.20

```
billableMiles = max(12.00, 3) = 12.00

Distance charge:
  slab 1–10   = 10 × 4.00 = 40.00
  slab 11–12  =  2 × 3.50 =  7.00
  distanceCharge = 47.00

waypointFee = 2 × 6.00 = 12.00
baseSubtotal = 47.00 + 8.00 + 12.00 + 12.00 = 79.00
surchargeTotal = 20.00
subtotal = 79.00 + 20.00 = 99.00
taxAmount = 99.00 × 0.20 = 19.80
total = 99.00 + 19.80 = 118.80
```

10. Return Journey Rule

A return journey is not calculated as one combined fare first. Each leg is quoted independently.

1. Calculate outbound leg using outbound miles, stops, pickup time, and selected options.
2. Calculate inbound leg using inbound miles, stops, pickup time, and selected options.
3. Total return price = outbound total + inbound total.

Screenshot Placeholder 3
Add screenshot of quote result / breakdown screen here.

11. Validation and Failure Conditions

- If no active state matches the pickup location, quote generation fails.
- If the resolved state has no active rates, quote generation fails.
- If a rate already exists for the same state + vehicle, a duplicate save is blocked.
- If slab JSON is invalid, save is blocked.
- If slab ranges have gaps or overlaps, save is blocked.
- If taxRate exceeds 0.30, save is blocked.
- If referenced state or vehicle is missing, save is blocked.

12. Current Implementation Notes

- Tax rate is stored as decimal, not percentage text. Example: 0.20 means 20%.
- Night surcharge is currently a flat fee, not a multiplier.
- Waypoint fee is currently charged per stop.
- Airport pickup fee is currently always added as part of the active rate formula.
- All quote totals are returned in GBP.

13. Source References

- `src/lib/quote-pricing.js`
- `src/lib/slab-utils.js`
- `src/models/StateVehicleRate.js`
- `src/app/api/admin/state-vehicle-rates/route.js`
- `src/app/api/admin/state-vehicle-rates/[id]/route.js`